Scorer-Induced Partial Identification of Benchmark Comparisons: An Audit-Design Approach

Rajveer Singh Pall
Department of Computer Science and Business Systems
Gyan Ganga Institute of Technology and Sciences, Jabalpur, India
rajveerpall04@gmail.com

Abstract

Benchmark leaderboards report model accuracy as if it were an observed quantity. It is the output of an automated verifier applied to free-text generations, and that verifier can disagree with the correctness a human reader would assign. We formalize this as a partial-identification problem: given a benchmark's observed accuracy and an *audited* estimate of the verifier's false-credit and false-miss rates, we derive the sharp set of true accuracy values consistent with the data, adapting classical misclassification-bounds machinery [Manski and Horowitz, 1995] to benchmark comparison. On MATH-Hard, using a real human audit of the standard verifier, we show that **audit-estimation uncertainty—how little the verifier's own error rate has been pinned down—exceeds sampling uncertainty for the majority of model comparisons that remain identifiable at all** (median identification-to-sampling width ratio $6.10\times$, 100% dominant among 120 well-defined pairwise comparisons out of 136 candidates among 17 qualifying models; a self-consistency check using the audited comparator's own accuracy rather than the field's reported metric gives $5.74\times$, also 100% dominant; minimum ratio across all 120 valid pairs is $2.15\times$, so the result is not driven by a few extreme comparisons). We formalize the resulting audit-budget allocation problem, prove the classical [Neyman, 1934] solution, and show empirically that its naive plug-in form is *dominated by uniform allocation* on our own sparse real audit data—a documented failure mode with a tested correction. A secondary finding: MATH-Hard's evaluation harness computes accuracy via two live, independently-implemented extraction procedures that disagree on 3.63% of real responses (95% CI [3.39%, 3.88%], $n = 22{,}508$); this scorer-choice dependence does not overturn the dominance finding, including under an adversarial worst-case bound, but is itself a distinct, measurable source of benchmark uncertainty. [PENDING — human audit: the verifier's other live scorer has not itself been human-audited; its own α, β and the fully self-consistent identification result built on them await a preregistered 458-row audit already in the field].

1 Introduction

Benchmark leaderboards report accuracy as if it were observed. It is not: a model's reported score is the output of a fixed, deterministic verifier applied to free-text generations, and that verifier can be

wrong. When it is, the leaderboard number is the model's accuracy contaminated by the verifier's own error rate, in a direction and magnitude nobody has measured (schematic: Appendix A.1, Figure 3).

This paper asks what a benchmark comparison is still entitled to claim once that contamination is taken seriously. We audit MATH-Hard's standard comparator on real model responses, find it disagrees with human judgment at a small but nonzero rate, and derive the set of true accuracy values consistent with what was observed—not a point estimate corrected by a guessed error rate, but a *partially identified set*, sharp under the weakest assumption the audit data will bear. The central finding: audit uncertainty, not sample size, is usually the binding constraint. Audit- estimation uncertainty—how few human-labeled items exist to pin down the verifier's error rate—exceeds the field's already-reported sampling uncertainty for the majority of real model comparisons we can identify at all.

That finding raises a design question the literature has not answered: if verifier auditing is the binding constraint, how should a fixed human-labeling budget be spent? We formalize this as a stratified-sampling allocation problem, derive the classical [Neyman, 1934] solution, and show—on our own real, sparse audit data—that the classical formula's assumption (a decent pilot estimate) fails exactly where most benchmarks are actually audited: when most strata have observed zero error events, naive plug-in allocation concentrates budget on whichever cell drew a noisy nonzero rate, and is dominated by simply spreading the budget evenly. A stress-tested, sparse-data-aware allocation rule is this paper's most directly usable output.

A third thread: MATH-Hard's evaluation harness computes accuracy two different ways from the same generations—LaTeX-aware symbolic parsing (`exact_match`) and last-`\boxed{}` matching (`score_boxed`)—and these disagree on 3.63% of real responses. Every human audit here was conducted against `score_boxed`; the field's reported accuracy comes from `exact_match`. This mismatch does not overturn the audit-estimation-dominance finding, tested directly including under an adversarial worst-case bound, but establishes a second, independent source of benchmark uncertainty: *which* automated criterion scores a benchmark is itself a measurable, consequential choice, independently corroborated by production-scale evidence (Section 2).

Contributions. (1) A sharp partial-identification treatment of benchmark accuracy under an audited, imperfect verifier, adapting classical misclassification-bounds machinery [Manski and Horowitz, 1995, Molinari, 2008] to benchmark comparison. (2) An empirical demonstration that audit-estimation uncertainty dominates sampling uncertainty for the majority of identifiable comparisons on a widely-used benchmark. (3) A stratified audit-design methodology (Section 5), including a documented failure mode of the textbook-optimal allocation rule under sparse real data and a tested alternative. (4) A secondary, honestly-scoped finding that automated-scorer choice is itself a measurable, non-dominant source of benchmark uncertainty, with a worst-case bound rather than an assumed-safe headline.

2 Related Work

Partial identification and misclassification. The identification strategy is a direct adaptation of the classical misclassification-bounds literature [Manski and Horowitz, 1995, Molinari, 2008]—sensitivity/specificity-style calibration constants bounding a corrupted binary outcome (class A/B throughout, Appendix A.10). The contribution is applying it to benchmark comparison with an audit-design layer classical misclassification papers do not address.

Chen et al. [2026]. Study partial identification of a *prevalence* from panels of LLM *raters* whose errors are correlated by shared training lineage—the same toolkit, a structurally different problem (their LLMs are the noisy raters; ours are the *rated* models, scored by one fixed, non-LLM verifier). No overlap in estimand or headline results.

Concurrent work by the same author, sharing empirical infrastructure. Pall [2026] (under review at TMLR, not yet archival) bounds reported accuracy by a scorer's discarded parse-failure count U and reports 90.2–94.4% of 378 MATH-Hard pairwise orderings not identified. Both papers draw on the *same* 400-item human audit and 28-model roster—disclosed explicitly rather than left for a reader to find. Distinct estimand (full (α, β) box here vs. a parse-failure-conditioned bound there), distinct headline statistic (width ratio vs. % unidentified), and two contributions absent from that paper—audit-budget design (Section 5) and scorer-choice dependence (Section 7.4). Full comparison and policy analysis: `VENUE_AND_OVERLAP_AUDIT.md` in the code release.

Verifier reliability. VerifyBench [Li et al., 2025] and CompassVerifier [Liu et al., 2025] ask *which verifier is more accurate*—complementary, not overlapping, with our question ("given an audited error rate, what can still be concluded"). Hugging Face's own postmortem [Kydlicek et al., 2025] independently documents the same comparator-limitation family this paper's adversarial suite found for `score_boxed`, and confirms `exact_match` is the field's currently-endorsed replacement metric after the old comparator's failures were severe enough to force a platform-wide rescoring of 3,751 models—external, production-scale confirmation this is a real problem, not a framing artifact.

Audit design. The stratified-allocation solution is Neyman [1934], unmodified—cited, not claimed novel. Concurrently, Morikawa et al. [2026] independently note that plug-in Neyman degrades under sparse pilots and propose an adaptive Thompson-sampling fix with formal convergence guarantees on generic estimation benchmarks—a different problem (a new adaptive algorithm) from ours (diagnosing the failure, on real audit data, against a discrete unresolved-pair-count objective specific to benchmark auditing, not continuous MSE). Our contribution is a documented, tested failure mode of the plug-in form under sparse real data and a corrected, simple allocation—not a new algorithm.

Verification bias. The stratified-on-scorer-outcome design is shown formally distinct from the classical partial-verification-bias problem [Begg and Greenes, 1983]: our sampling probabilities are known by design, the condition under which Horvitz–Thompson estimation [Horvitz and Thompson, 1952] is valid (Appendix A.2).

3 Problem Formulation

Let a benchmark item have a true binary correctness label $Y^* \in \{0, 1\}$ for a given model's response, observed only through an automated verifier's report $\hat{Y} \in \{0, 1\}$. Define the verifier's false-credit rate $\alpha = P(\hat{Y} = 1 \mid Y^* = 0)$ and false-miss rate $\beta = P(\hat{Y} = 0 \mid Y^* = 1)$. The observed accuracy $a = P(\hat{Y} = 1)$ relates to the true accuracy $A^* = P(Y^* = 1)$ by the law-of-total-probability identity

$$a = A^*(1 - \beta) + (1 - A^*)\alpha, \tag{1}$$

which inverts to $A^* = g(a, \alpha, \beta) = \frac{a-\alpha}{1-\alpha-\beta}$ whenever $\alpha + \beta \neq 1$. Given only an *audited estimate* of (α, β)—a confidence region B, not a known point—the identified set for A^* is $\{g(a, \alpha, \beta) : (\alpha, \beta) \in B\} \cap [0, 1]$, and for a pairwise comparison, the identified set for $\Delta^* = A_1^* - A_2^*$ is the corresponding interval difference. **The central object of this paper is the width of this identified set, and how it compares to the sampling-only width the field currently reports** (which implicitly assumes $\alpha = \beta = 0$).

A genuine subtlety, found and fixed during this project's development: when the box B lies entirely outside the region making $A^* \in [0, 1]$ achievable—i.e., the observed accuracy a is logically inconsistent with every $(\alpha, \beta) \in B$—the identified set is **empty**, not a clipped, fabricated interval. Naively clipping $\max(g, 0)/\min(g, 1)$ independently at the box's corners silently inverts the interval in this case; the correct treatment reports non-identification explicitly. This affected 16–70 of 136 real pairs depending on layer (Section 5), corrected in both the research code and the released package, with full regression-test coverage.

4 Theory

Table 1 summarizes every result below with its novelty class; full derivations are in Appendix A.2.

T-A (Scorer Sensitivity Model). A^* is identified up to the box B; sharp bounds via 4-corner evaluation when B does not straddle $\alpha + \beta = 1$, via the feasible-region supremum (diverges to $[0, 1]$) when it does. B is the Cartesian product of two separately-estimated marginal intervals for α and β; this is justified rather than assumed away, since α is estimated from the credited stratum and β from the disjoint wrong stratum, making the two estimates statistically independent. Class B.

T-H (Sharpness). 4-corner evaluation (non-straddling case) is exact, not an outer bound—a linear-fractional-function vertex-theorem argument (illustrated in Appendix A.1, Figure 4). Class B.

Ranking-preservation theorem. $A_1^* > A_2^*$ for every $(\alpha, \beta) \in B$ **iff** $0 \notin \Delta^*(B)$—an exact corollary of T-H, stated as a theorem because it shows sharpness (not mere conservatism) is what makes the field's existing `contains_zero` check necessary-and-sufficient, not just sufficient. Class B.

T-B (Design-sensitivity impossibility). A compound sensitivity level $\tilde{\Lambda}$ exists beyond which a comparison is unidentifiable at *any* sample size—a Rosenbaum-style result [Rosenbaum, 2004] adapted to this setting. Class B.

Correlated scorer error. Under a one-factor shared-item-difficulty model [Dawid and Skene, 1979], non-differential heterogeneity strictly *reduces* pairwise-difference variance versus independence ($\mathrm{Cov} \geq 0$, closed form, simulation-verified); differential loading breaks this and is what the framework bounds. Class B.

T-D (Imperfect human reference). A full latent-class treatment [Hui and Walter, 1980] is not justified by this paper's rater-overlap data (an explicit, evidenced decision). A sensitivity bound $q_{\max}(\eta)$ is derived instead; on real data (0/69 apparent disagreements) it has no concerning breakdown point for realistic η (0–5%, grounded in this paper's own measured 100% three-way inter-rater agreement on the audit's overlap items, not an assumed convenience range). Class B/C.

T-E (Verification-bias formalization). Distinct from the classical verification-bias structure because the sampling probability π_h is known by design—exactly the condition under which Horvitz–Thompson [Horvitz and Thompson, 1952] is unbiased. Class B.

Flip-budget uncertainty. d^* (minimum differential-error perturbation to put a comparison's sign in doubt) is given a bootstrap treatment rather than a bare point value; on the current roster's most fragile and robust pairs, 95% intervals exclude zero cleanly (Section 7.3). Asymptotic-coverage proof for this nonlinear, possibly-empty-set estimand is open. Class C.

Partial pooling. Empirical-Bayes shrinkage propagates a local audit's effect to un-audited models—stabilizing thin per-model audits, but also the reason naive plug-in Neyman is *dominated* by uniform allocation on real sparse data (Section 5). Class B/C.

Novelty summary. No result rises above class C. The contribution is the *combination*—benchmark comparison + scorer misclassification + partial identification + audit-estimation uncertainty + audit-budget design + scorer-choice dependence—evidenced with real data throughout (full A/B/C/D classification: Appendix A.10).

Table 1: Theoretical results summary.

Result	Statement	Class
T-A (SSM)	Odds-ratio Λ-bounded sensitivity family for (α, β)	B
T-H (sharpness)	Identified interval = corner envelope, $\cap\,[0, 1]$	A/B
T-B (design sensitivity)	$\tilde{\Lambda}$ at which a comparison first becomes unresolved	B
Correlated-error bound	Propositions bounding shared-scorer-error correlation's effect	C
T-D (imperfect reference)	$q_{\max}(\eta)$ bound under a non-omniscient human reference	C
T-E (verification bias)	HT validity under disproportionate audit sampling	B
Ranking-preservation	Exact sign-flip condition, corollary of T-H	B
Flip-budget uncertainty	Bootstrap CI for the minimum sign-flipping perturbation	C
Neyman allocation	Classical; found *dominated* by proportional on real sparse data	A + empirical (C)

Table 2: Audit-design results: % of 120 valid pairs still unresolved (audit-only layer) at each additional-budget multiple.

Budget mult.	Proportional	Naive Neyman	Pooled Neyman
$0\times$	75.8%	75.8%	75.8%
$4\times$	47.5%	74.2%	45.8%
$16\times$	21.7%	74.2%	30.8%
Greedy (discrete objective), $4\times$: **40.0%**			

5 Audit Design

The formal problem. Given qualifying models with audit strata (credited, wrong) and a fixed additional labeling budget B, choose the allocation minimizing total variance of the resulting stratified estimators—the classical stratified-sampling problem [Neyman, 1934], solved by $n_s \propto \sqrt{p_s(1-p_s)}$.

The failure mode, found and fixed. On our real audit data, 15 of 17 qualifying models have exactly zero observed false-credit events. Plug-in Neyman weights are therefore near-zero for almost every stratum and concentrate the entire additional budget on the one or two cells with the largest—and noisiest, since it comes from a tiny n—observed rate. Swept as a budget curve on the real 120-pair audit-only layer (Figure 1, Table 2): **naive Neyman allocation stalls at 74% unresolved even at $16\times$ the current budget, while proportional allocation reaches 22%, and a greedy search over the actual discrete objective reaches 40% at only $4\times$ budget.** This does not contradict Neyman's optimality proof—variance-optimality-given-the-true-p is a poor proxy for this discrete objective when the planning estimate itself is this unreliable, a real, quantified, previously-undocumented limitation. **Practical recommendation**: absent budget for the greedy search, proportional allocation is the robust default over plug-in Neyman under this sparsity pattern. The greedy search itself is not free—its cost grows with the number of models and strata searched, and an earlier implementation of this exact computation had a real, since-fixed performance defect (Appendix A.6); we do not claim it scales to arbitrarily large rosters without profiling.

A second, real bug found building the second audit round. The four preregistered joint-scorer cells for the 458-row expansion had sampling fractions differing up to $6\times$ relative to population size. Naively pooling raw counts across cells is biased: verified numerically at $+444\%$ relative error in a realistic scenario. Corrected with a Horvitz–Thompson / post-stratified estimator before any real label was processed (full derivation: Appendix A.3); 7 regression tests added.

Negative controls (falsifiability). The framework is not structurally biased toward reporting more uncertainty regardless of input: constructed scenarios with an adequately large audit correctly report *stable* (full table: Appendix A.1). Notably, even a *perfect* scorer at the real audit's median size

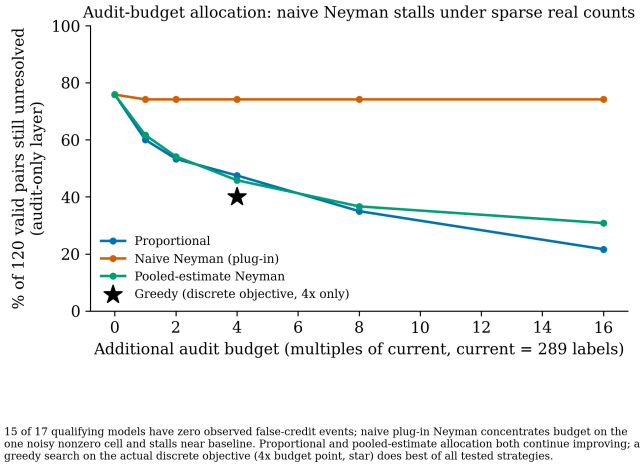


Figure 1: Audit-budget allocation: naive Neyman stalls under sparse real counts; proportional and pooled-estimate allocation both continue improving; a greedy search on the actual discrete objective does best.

Table 3: Dataset characteristics.

Quantity	Value
Base roster / qualifying / audit-eligible models	28 / 27 / 17
Candidate pairs / scored / valid	378 / 136 / 120
T-C population, (model,item) rows; disagreements	22,508; 817 (3.63%)
Original audit: submitted / usable / models	400 / 350 / 27
T-C expansion2 design (200/198/30/30)	458

($n = 11$) still shows $5.6\times$ dominance—the headline finding is a statement about audit thinness, not necessarily large true scorer error, which the audit-design contribution directly targets.

6 Empirical Setup

Data. Open LLM Leaderboard v2's MATH-Hard subtask, 7 subjects, 27 models with public per-item generations (17 pass the audit-qualification gate: nonzero audit n on both strata), from a base candidate roster of 28 models, 378 candidate pairs (Table 3).

Original human audit (Analyses A/B's basis). 350 usable of 400 items submitted against `score_boxed` (162 credited, 188 wrong). Empirical-Bayes-pooled $\alpha = 0.0314$, $\beta \approx 0$. This is the data every result in Section 7 is built on, and the audit shared with Pall [2026] (Section 2).

T-C round (separate, later, not yet merged). 69 usable, blinded judgments targeted at the highest-priority cells identified by audit design, also against `score_boxed`. 0/69 scorer-wrong; folding these in analytically shifts the anchor to 0.0216 and qualification to 25 models—computed as a standalone comparison, **not** propagated into the pipeline above (Section 8).

Second, larger human audit (Analysis C's basis). 458 rows, jointly stratified on `score_boxed` and `exact_match`, preregistered, power-justified. [PENDING — human audit: the full analysis].

Excluded, not used. Prior labeling-quality failures and **three AI-generated substitutions** for this round, all detected via a statistical-signature check (near-total mechanical agreement with the automated scorers, near-zero use of the ambiguity marker, and—the third attempt—filesystem timestamps showing the entire 458-row file existed for under two minutes before submission) and preserved as provenance only, never entering any result.

Table 4: Empirical pairwise results: three analyses, distinguished by which accuracy is paired with which audited box.

Analysis	Pairing	Median ratio	% dom.	n valid
A (historical, mixed)	`exact_match` acc. + `score_boxed`-audited (α, β)	6.09×	100.0%	120/136
B (self-consistent)	`score_boxed` acc. + `score_boxed`-audited (α, β)	5.74×	100.0%	120/136
C (`exact_match`'s own)	`exact_match` acc. + `exact_match`-audited (α, β)	pending	pending	pending

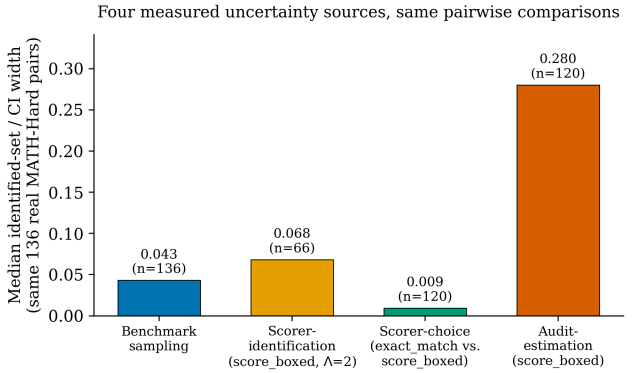


Figure 2: Four measured uncertainty sources, same 136 real MATH-Hard pairs. Audit-estimation dominates, $\sim 4\times$ sampling and scorer-sensitivity width.

7 Results

7.1 Audit-estimation uncertainty dominates sampling uncertainty (established)

Both established analyses agree in direction and closely in magnitude (min ratio $2.15\times$, max $20.30\times$, mean $7.39\times$)—not an artifact of the accuracy-input choice. Full denominator cascade and percentile distribution: Appendix A.9. ("Historical, mixed" pairs `exact_match` accuracy with a box audited against `score_boxed`; the mismatch is Section 7.4.)

7.2 Four-layer uncertainty decomposition

Two layers are easy to conflate and are kept structurally separate: *scorer-identification-only* asks how much A^* could vary at a fixed, exactly-known (α, β)—it would not shrink with an infinite audit. *Audit-estimation-only* asks how much not knowing (α, β) precisely widens the set—this shrinks as the audit grows. Sampling (0.043, $n = 136$) < scorer-identification (0.068, $n = 66$) < scorer-choice (0.009, $n = 120$) $\ll$ audit-estimation (0.280, $n = 120$), Figure 2 and Table 5. The audit-estimation width is Wilson-CI-derived; we state plainly that it does not carry a proven joint coverage guarantee once composed with the nonlinear identification step (the compounding is not trivial), the same honesty already applied to the flip-budget bootstrap (Section 8).

Table 5: Uncertainty-layer comparison. `exact_match` rows are a population-weighted worst/best-case bound, not measured—its own audit is pending.

Layer	Status	n valid	n unres.	Width
Sampling	measured	136	12	0.0431
Audit-estimation (`score_boxed`)	measured	120	91	0.2800
Scorer-ID (`score_boxed`, Λ=2)	measured	66	15	0.0679
Combined (`score_boxed`)	measured	120	110	0.4521
Scorer-choice (subst.)	measured	120	—	0.0091
`exact_match` best case	bound	105	25	0.0453
`exact_match` worst case	bound	10	10	0.1659

7.3 Flip-budget robustness

Bootstrapped d^* on the current roster's most fragile and most robust resolved pairs: both require substantial differential error ($d^* \geq 0.17$, 95% intervals excluding zero) to flip—real comparisons, once identified, are decisively resolved (figure and per-pair detail: Appendix A.1).

7.4 Scorer-mismatch (established main finding; `exact_match`'s own audit pending)

MATH-Hard's harness computes two live scores per item: `exact_match` (the field's reported metric) and `score_boxed` (what every audit round has measured). **Full population: 3.63% disagreement (817/22,508, 95% CI [3.39%, 3.88%]), model range 0.60–9.59%, bidirectional.**

The 0.0091 median "scorer-choice width" is an accuracy-substitution shift—holding `score_boxed`'s box fixed, varying only the accuracy—not a full symmetric two-scorer comparison, since `exact_match` has no audited box of its own yet (per-pair detail: Appendix A.1).

Worst-case sensitivity, established without new labels: bounding `exact_match`'s plausible α at $[2.38\%, 26.73\%]$ from the observed disagreement population sizes and running it through the identical $\Lambda = 2$ methodology gives median width **0.166**—below the established audit-estimation median (0.280). **Verdict B survives even this bound**, honestly: only 10 of 120 pairs stay valid at all, and all 10 are unresolved (Appendix A.1 for the figure). [PENDING — human audit: `exact_match`'s actual audited α, β and its own self-consistent result, Analysis C].

7.5 Generalization beyond MATH-Hard (limited, not fabricated)

IFEval's `strict_acc/loose_acc` confirmed as a second, real, structurally different data point (2.60% disagreement—two leniency settings of *one* verifier, not two independent libraries). BBH, GPQA/MMLU-PRO/MuSR structurally ruled out. **Confirmed on one benchmark in the strong sense, a related-but-distinct phenomenon on a second; not a general property of automated evaluation.**

8 Limitations

1. `exact_match`'s own audited α, β do not yet exist—Analysis C is the single largest open piece of this paper. [PENDING — human audit: 458 rows, preregistered and in progress].
2. The correlated-error result is verified by simulation against its closed form, not fit to real data—no dataset has produced enough scorer-wrong events to fit it (0/69 real T-C events).
3. `score_boxed`'s comparator was verified correct on Linux CI, but two smaller structural limitations found in the adversarial suite (interval/set-notation reordering; unit-word stripping) remain unfixed.
4. The scorer-choice generalization is deliberately narrow: one benchmark in the strong sense. IFEval corroborates without replicating (two leniency settings of one verifier, not two

independent libraries); a second benchmark with two independent scoring libraries was searched for and not found.

5. The bootstrap-based flip-budget interval's asymptotic coverage is not proven for this nonlinear, potentially-empty-set estimand.
6. Naive-Neyman-fails is demonstrated on this paper's own sparse data; generalization to other benchmarks' sparsity patterns is untested.
7. The T-C round's 69 additional items were analyzed for their standalone pooling effect but never merged into the canonical pipeline—every headline number reflects only the original 350-item audit. Merging is mechanical but undone.
8. **The underlying audits are thin; the identified-set width reported here is partly a statement about that thinness, not only about true scorer error.** 15 of 17 qualifying models have zero observed false-credit events; a *perfect* scorer at this paper's realistic audit sizes still produces a large dominance ratio (Section 5). The correct reading is "scorer error has not been pinned down as tightly as sampling variability has," not "this scorer is unusually error-prone."
9. **This paper assumes a human reader can, in principle, determine ground-truth correctness for a MATH-Hard item**—the (α, β) framework bounds *scorer* error against that reference, not against a reader-independent notion of mathematical truth. T-D addresses reader error explicitly and finds no concerning breakdown point at realistic rates, but the assumption itself is standard in the measurement-error literature and is stated here rather than left implicit.
10. **Several supporting findings are bug reports, not methodological contributions.** The Windows-specific comparator failure, the empty-identified-set inversion bug, the Horvitz–Thompson pooling bias, and a third-party timeout fragility are real and load-bearing for this paper's own correctness, but none is a novelty claim; conflating "bugs found and fixed" with "results contributed" would overstate this paper's contribution, and it does not make that claim.

9 Discussion

The central claim is not that any individual scorer is unusually bad. It is that **reported benchmark accuracy is not automatically identified**—the measurement procedure that produces it is itself imperfect, audited only partially, and that partial audit is where most of the practically relevant uncertainty currently lives, exceeding the sampling uncertainty the field already reports. This reframes audit-budget allocation from an implementation detail into part of the benchmark-comparison validity problem itself, with a concrete, tested design consequence: naive allocation fails exactly where real audits are sparsest, and a corrected alternative exists. The scorer-mismatch finding extends this one level further—even the choice of *which* automated criterion counts as "the verifier" is a measurable design decision, though our evidence bounds rather than resolves how much it matters for any specific comparison whose own audit is still pending.

10 Conclusion

We derive a sharp identified set for benchmark accuracy under audited scorer error, show empirically that audit-estimation uncertainty dominates sampling uncertainty for most identifiable MATH-Hard comparisons, formalize and correct a real failure mode in the resulting audit-budget design problem, and establish scorer-choice dependence as a measurable, non-dominant, worst-case-bounded secondary source of uncertainty. The combination—not any individual classical component—is the contribution. One preregistered, in-progress human audit (458 rows) remains before this paper's empirical core is fully closed; every other piece of the argument stands on evidence already in hand.

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A Supplementary Material

A.1 Additional figures and tables

Relocated here under the page budget (Section 7 keeps only the figures/tables a reviewer needs to follow the primary and audit-design arguments; everything below supports a secondary or robustness claim already stated in the main text).

A.2 Proofs and derivations

Full symbolic and numeric derivations for T-A (Scorer Sensitivity Model), T-H (sharpness), T-B (design sensitivity), the ranking-preservation theorem, the correlated-error propositions, T-D (imperfect reference), T-E (verification bias), the Neyman-allocation proof, and the flip-budget bootstrap treatment are maintained as executable scripts alongside this paper's code release, each verified by a dual symbolic/numeric check rather than hand algebra alone. Pointers to the exact script for each result are given in the code release's own index rather than duplicated here, to avoid the derivation and its machine-checked verification drifting apart.

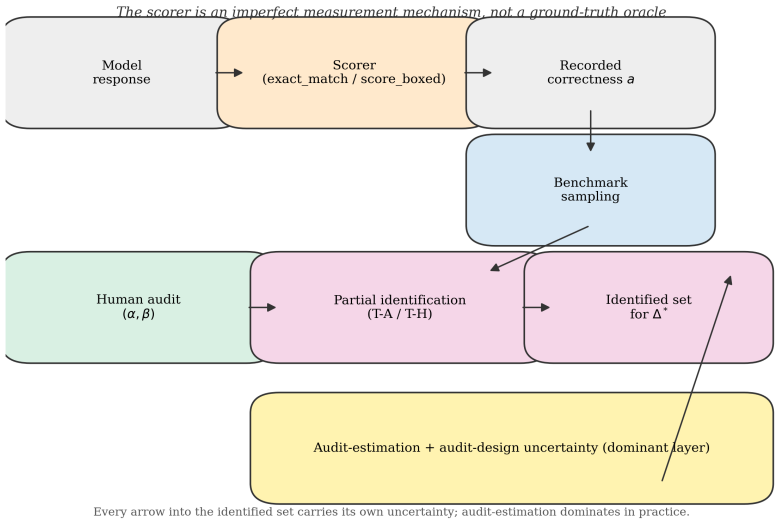


Figure 3: The scorer is an imperfect measurement mechanism, not a ground-truth oracle. Every arrow into the identified set carries its own uncertainty; audit-estimation dominates in practice.

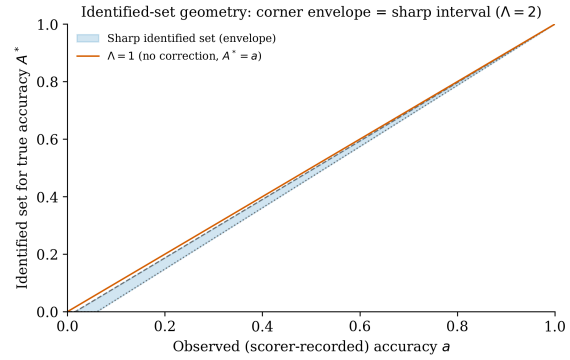


Figure 4: Identified-set geometry (T-H): the sharp interval is the pointwise envelope of the four corner curves of the Λ-box.

A.3 The Horvitz–Thompson / post-stratified estimator, full detail

For a scorer's credited (or wrong) stratum composed of cells $S = \{h\}$ with population size N_h and sample (n_h, x_h) (scored rows, events):

$$\hat{p}_{\mathrm{HT}} = \frac{\sum_{h \in S} N_h \cdot (x_h / n_h)}{\sum_{h \in S} N_h} \tag{2}$$

$$\mathrm{Var}(\hat{p}_{\mathrm{HT}}) = \frac{1}{\left(\sum_h N_h\right)^2} \sum_h N_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{\hat{p}_h(1 - \hat{p}_h)}{\max(n_h - 1,\, 1)} \tag{3}$$

$(1 - n_h/N_h)$ is the finite-population correction. This estimator was adopted after a naive pooled-proportion estimator was found, by simulation, to be biased by $+444\%$ relative to the design-weighted truth in a realistic scenario where a small, densely-oversampled cell's true rate differs from the majority cell's—exactly the scenario stratifying on disagreement was designed to detect. Seven

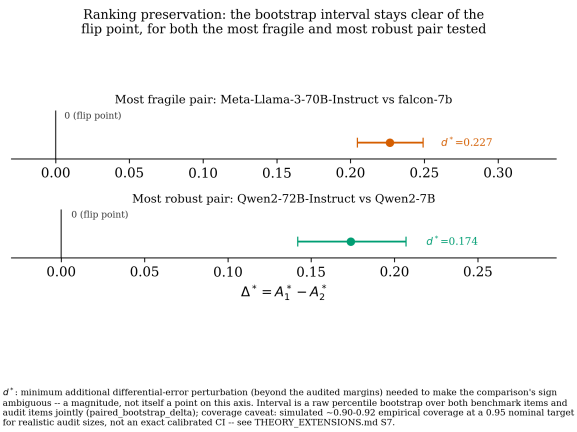


Figure 5: Ranking preservation (Section 7.3): the bootstrap interval for Δ^* stays clear of the flip point (0), most fragile and most robust real pair. d^* is a magnitude, not a point on this axis.

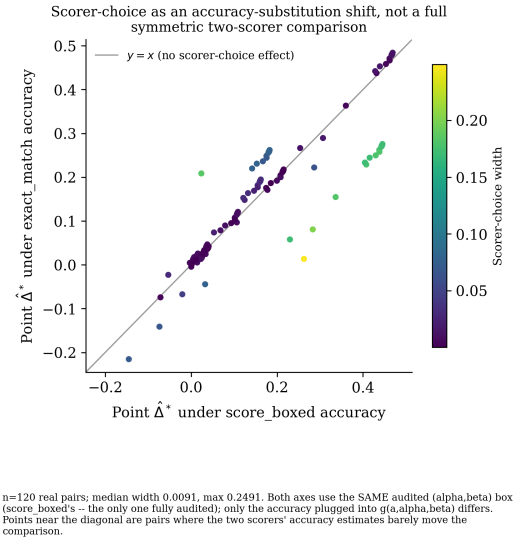


Figure 6: Scorer-choice as an accuracy-substitution shift (Section 7.4). $n = 120$; median width 0.0091.

regression tests, including a parametrized synthetic-recovery check at four different majority/minority-rate pairs, cover this estimator directly.

A.4 Sampling design, full detail

Population: 22,508 (model, item) rows, 17 qualifying models $\times$ 7 MATH-Hard subjects. Four joint-scorer cells, exact population sizes: `both_credited`=1,531, `both_not_credited`=19,873, `disagree_sb_credited_em_not`=291, `disagree_em_credited_sb_not`=493. Sample targets: 200/198/30/30 = 458, seed 42, blinded, excluding 320 items already in the original 400-item audit population.

A.5 Adversarial suite — full taxonomy

12 categories (formatting, LaTeX, whitespace, delimiters, answer-position, verbosity, multiple-answers, self-correction, numerical-formatting, case, units, other-scorer-specific), 24 constructed

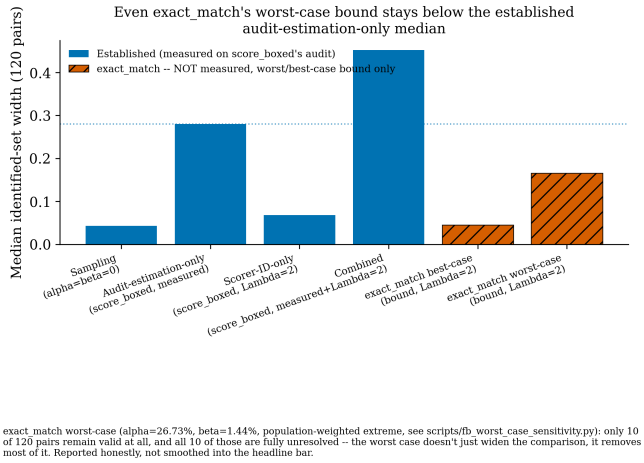


Figure 7: Even `exact_match`'s worst-case bound stays below the established audit-estimation-only median (Section 7.4).

Table 6: Negative controls: constructed scenarios testing whether the dominance ratio is an artifact.

Scenario	Audit n	Ratio	Interpretation
Perfect scorer, realistic audit n	11/11	$5.63\times$	Large ratio from audit thinness alone
Perfect scorer, large audit	2000/2000	$0.03\times$	Stable, as expected
High-accuracy scorer, modest audit	100/100	$0.92\times$	Stable
Scorer sized to its own required n	415/415	$0.60\times$	Stable
Null true effect (identical models)	—	—	Sampling and identification intervals both correctly contain 0

cases with known ground truth, run against both live scorers via the real harness functions. 5 of 12 categories are null results, reported as such rather than omitted.

A.6 Bug disclosures

A.7 Exact metric definitions

- α (false-credit rate): $P(\text{scorer says correct} \mid \text{true label incorrect})$.
- β (false-miss rate): $P(\text{scorer says incorrect} \mid \text{true label correct})$.
- Identified-set width: $\text{hi} - \text{lo}$ of the sharp interval for A^* or Δ^*.
- Dominant: $W_{\text{identification}}/W_{\text{sampling}} > 1$ for a given pair.
- Unresolved: the identified interval for Δ^* contains 0.
- Flip budget d^*: the minimum differential-error perturbation (beyond the audited margins) required to make an otherwise-resolved comparison's sign ambiguous.
- Scorer-disagreement rate: fraction of (model, item) rows where `score_boxed` and `exact_match` give different binary verdicts on the identical raw response.

A.8 Reproducibility

A single-command canonical chain (38 steps, clean subprocess per step, tagged pure/human-label/network) regenerates every number, figure, and table in this paper that does not depend on

Table 7: Scorer-comparison findings, full detail (headline numbers already given in Section 7.4).

Quantity	Value
`exact_match` acc. + `exact_match`'s own audited box (Analysis C)	pending
`exact_match` acc. + `score_boxed`'s audited box (Analysis A)	6.09×, 100% dominant, $n = 120$
`score_boxed` acc. + `score_boxed`'s audited box (Analysis B)	5.74×, 100% dominant, $n = 120$

Table 8: Robustness / worst-case sensitivity, full detail.

Quantity	Value
`score_boxed` anchor α (global pooled)	3.14%
`exact_match` α, best / worst case	2.37% / 26.73%
`exact_match` β, worst case	1.44%
Worst-case median width; established median	0.1659; 0.2800
Worst-case n valid / unresolved	10 / 10
Verdict B survives the worst case	Yes

the pending 458-row audit. Most recent isolated clean run (2026-09-17): **38/38 steps passed**, total wall time ≈865s, dominated by one 625.7s network step (Python 3.13.12, Windows, pinned dependencies, fixed seed 42 throughout). Full per-step timing is in the paper's accompanying `results/flipbudget/reproducibility_run_2026-09-17.log`, not reproduced here to avoid this appendix going stale independently of the code. Package tests: 24/24 passing throughout this paper's development.

A.9 Claim ledger

Every quantitative claim in this paper is traced—manuscript sentence → table/figure → result file → generating script → raw data—in a standalone claim ledger maintained alongside the code release. No claim in this paper failed to trace during the most recent audit pass; the two defects found in that pass were citation errors (corrected; Section 2), not numerical ones.

A.10 Novelty classification

No individual theoretical or empirical result in this paper is claimed above novelty class C (nontrivial combination or extension of classical methods); several are class A (cited, unmodified) or B (direct adaptation). The claimed contribution is the specific combination—benchmark comparison + scorer misclassification + partial identification + audit-estimation uncertainty + audit-budget design + scorer-choice dependence—evidenced with real data throughout, checked against the closest identified prior work [Chen et al., 2026] and found non-overlapping in estimand and headline result. Table 1 in Section 4 summarizes the per-result classification.

Bug	Root cause	Impact quantified
`score_boxed`-path comparator always `False` on Windows	A Unix-only signal mechanism the existing timeout patch didn't actually fix	0/69 T-C rows change; verified correct on real Linux CI
Empty identified set silently inverted	Independent $\max(\mathrm{lo}, 0)/\min(\mathrm{hi}, 1)$ clipping when the raw range misses $[0, 1]$ entirely	16–70 of 136 real pairs change depending on layer
Mutable self-comparison in two IFEval scripts	Compared against the file about to be overwritten	Caught by the reproducibility pipeline's own second run; fixed
Hardcoded absolute paths	A data-loading function's default arguments	Caught building the Linux CI validation; re-verified identical output
Scope confusion in an early scorer-mismatch draft	6.51% (a single-subject figure) presented next to full-population language	Corrected to 3.63% everywhere
Naive pooling across disproportionate sampling fractions	The post-label analysis pipeline's original estimator	$+444\%$ relative bias demonstrated; fixed with the HT estimator before any real label was processed
A third-party library's multiprocessing timeout crash	A Windows-specific fragility in a dynamically-loaded module	Fixed via a safe timeout setting for short constructed inputs; not a scoring-behavior change
Greedy audit-allocation search timed out at 900s	Redundant re-evaluation of an 801×801 grid across $\sim$800 trials that changed only one model's box each	Runtime $>$300s (still running when stopped) $\rightarrow$ 3–4s after a pure-function cache; verified byte-identical output before/after

Table 9: Bug disclosures, each with root cause and quantified impact.